Engineering Solution for Wind Turbine Telemetry Monitoring

Findings

The telemetry data was analysed using a python script and pandas library. The script grouped data by Turbine ID and calculated average temperature, maximum vibration level, and average RPM. Turbines were flagged only when the average temperature exceeded 85°C and maximum vibration level exceeded 15 mm/s.

Proposed Architecture

Wind turbines generate real-time telemetry data (Temperature, Vibration, and RPM). AWS Kinesis acts as a message queue to ingest and buffer incoming data streams. AWS Lambda processes telemetry events in real-time without requiring dedicated servers. Python anomaly detection logic evaluates incoming sensor data against standard maintenance thresholds. Amazon Timestream stores processed telemetry data and anomaly results. Grafana Dashboard visualises turbine performance and system health. Maintenance Alerts notify engineers when abnormal conditions are detected.

Cost Optimisation

For cost optimisation, store recent telemetry data in Timestream for fast querying and monitoring. Automatically archive older telemetry data to Amazon S3 after a defined retention period. Use AWS Lambda instead of dedicated servers to reduce infrastructure costs. Implement data lifecycle policies to move infrequently accessed data to lower-cost storage tiers. Using a serverless architecture ensures that the company only pays for the resources when being used. Balance performance and cost by separating frequently accessed data from archives.